



The Combinatorial Repertoire of Consciousness

Context-Dependent Gating, Temporal Basin Compression, and Recursive Access



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A finite nervous system supports an immense range of possible conscious contents. The Combinatorial Repertoire of Consciousness traces that range to a combinatorial repertoire of relatively stable local gates whose activation depends jointly on current activity, incoming signals, signal conjunctions, oscillatory phase, retained history, and the large-scale organization carried forward from the preceding transition. Conscious content corresponds to the distributed route taken through this gate space, together with the retained history and macrostate that remain causally active across the episode. A fully specified toy simulation makes the logic measurable, and the problem of specific conscious content gains an empirical form.

How a finite brain holds so much

How can a finite nervous system support an immense range of specific conscious contents?

How it works

1. **A combinatorial repertoire of gates.** The range can arise through relatively stable local gates whose activation depends jointly on current neural activity, external or afferent signals, signal conjunctions, oscillatory phase, retained history, and the large-scale organization carried forward from the preceding transition.
2. **Reuse multiplies possibilities.** Each local element can participate in many different episodes, each signal can enter many different conjunctions, and the joint pattern of active gates can route activity through a vast space of possible trajectories.
3. **Content as a route.** Conscious content is identified with the distributed route taken through this gate space, together with the retained history and macrostate that remain causally active across the episode.
4. **Combination joins history and re-entry.** Combinatorial coding joins with temporal basin compression, active inheritance, recursive access, and macrostate causal re-entry.
5. **Sequential gating shapes a geometry.** Sequential gating creates a finite-horizon state-space geometry. Broad sets of starting states can converge into reproducible endpoint regions, neighboring trajectories can separate across gate boundaries, and prior activity can reshape the transition map encountered by later activity.
6. **History is physical.** Retained history is physically instantiated in variables such as synaptic efficacy, adaptation, excitability, recurrent activity, neuromodulatory state, working-memory activity, or other persistent carriers.
7. **The macrostate feeds back.** A stabilized macrostate can feed back into subsequent gate conditions, so the larger organization produced by local activity participates causally in the local transitions that continue the episode.
8. **The conscious regime.** Conscious organization emerges when a large combinatorial repertoire is narrowed through sequential context-sensitive gating into a temporally stabilized, history-bearing, internally accessible, integrated macrostate whose organization causally re-enters the local dynamics producing the next moment.

9. **Lived experience.** Lived experience is identified with this accessible organization as lived from within its own continuing causal boundary. The dynamical mechanism and the phenomenal identity can be tested separately.

Terms to know

- **Gate.** A relatively stable local element whose activation depends jointly on several conditions at once. In plain terms: a switch that opens or stays shut depending on the whole situation, including what happened before.
- **Signal conjunction.** Two or more signals arriving together and acting as a combination.
- **Oscillatory phase.** Where an ongoing brain rhythm is in its cycle when a signal arrives.
- **Macrostate.** The large-scale organization of activity across the system.
- **Temporal basin compression.** Broad sets of starting states converging into reproducible endpoint regions. In plain terms: many different beginnings funnel into a few stable outcomes.
- **Macrostate causal re-entry.** The large-scale organization feeding back into the local gate conditions that continue the episode.
- **Recursive access.** Organization becoming available to the activity that produced it.

Precedents in biology

- **Adaptive immunity** shows how finite generative machinery can produce an enormous pre-encounter recognition repertoire through combinatorial assembly, junctional diversification, chain pairing, and selection, followed by selective expansion after cognate recognition.
- **Olfactory coding** gives a neural example in which distributed combinations of receptor responses distinguish a large stimulus space.
- **Candidate mechanisms** for conjunction-sensitive and context-dependent gating: mixed selectivity, dendritic nonlinearities, state-dependent computation, oscillatory phase sensitivity, recurrent population dynamics, and metastable neural states.

The model

Each gate receives an instantaneous feature map containing current state, input, and phase variables, together with pairwise conjunction terms, retained-history projections, phase-sensitive terms, and feedback from the preceding macrostate. The resulting gate vector contributes to macrostate selection, local state transition, history update, and subsequent phase evolution.

A candidate conscious regime is the conjunction

$$C_{org} = T \wedge H \wedge A \wedge K \wedge G$$

- T, temporal stabilization;
- H, consequential history;
- A, internal accessibility;
- K, macrostate causal re-entry;
- G, integration across multiple processes or systems.

The symbol $\wedge$ is logical AND: all five hold together. Each quantity is separately measurable, with distinct operational tests for each transition.

The numbers

A simple 24-gate toy simulation makes the combinatorial point measurable. These are model results under the declared toy parameterization:

- Fifty thousand sampled signal-context states generate 26,683 distinct 24-bit activation patterns from the same fixed gate architecture.
- Eight-step trajectories form four macrostate-centered endpoint groups with basin-conditioned RMS spread ratios of 0.0089, 0.0504, 0.2834, and 0.2516.
- Changing the retained-history state while holding the present-state grid fixed changes final basin assignment for 59.8% of grid points. In plain terms: with the present held the same, a different retained past changes where the system ends up for 59.8% of the present states tested.
- A recursive-feedback sweep produces 86.28% uninterrupted twenty-step macrostate persistence at feedback strength $\rho = 2.5$ and 100% at $\rho = 3.0$.
- A branched perturbation experiment starting from identical pre-perturbation states yields a 19.45% return probability with recursive macrostate re-entry active and 0.10% with the recursive term clamped. Final macrostate identity differs between branches for 53.575% of trajectories.

The simulation is a constructive demonstration of the operations and supplies exact numerical targets for reproduction and criticism.

A Period Lattice example

A low-dimensional Period Lattice example shows the combinatorial principle: four binary positions, sixteen ordered microstates, and five higher-order balance classes with multiplicities 1:4:6:4:1. Many local configurations participate in fewer higher-order organizational classes. At neural scale, the corresponding state space becomes far richer and temporally recursive.

The click of understanding

The click of understanding is a worked phenomenological case. Competing trajectories converge, a resolved organization becomes broadly accessible, and the stabilized state becomes available to the system as a unified relation.

What experiments can check

- Different experiences correspond to distinguishable distributions over gate paths, retained states, and macrostates across time.
- Tests can hold sensory information constant, or use ambiguous and matched-input conditions, and ask whether reported contents occupy separable route distributions beyond early sensory predictors.
- Controlled manipulation of phase, conjunction structure, retained-history carriers, basin geometry, and the physical realization of a macrostate can test successive parts of the mechanism.

The bigger picture

This is the third step at the level of specific contents. A deep present takes a particular route through a vast, history-sensitive gate space, and that route, together with the history and macrostate carried with it, is the content of the moment.

Full paper on PhilArchive: [The Combinatorial Repertoire of Consciousness: *Context-Dependent Gating, Temporal Basin Compression, and Recursive Access*](#)

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